Chapter 5

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Technologies of Harvesting and Field Handling Practices

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Farm management is generally aimed at maximizing the yield of a crop from the area of land under cultivation while at the same time maximizing the return. The time, labor and capital expended in bringing the crop to maturity is rewarded by the financial return obtained by the grower during marketing. For fruits and vegetables, the magnitude of this return mainly depends on: -quantity of fresh produce harvested and marketed, which mostly depends on production planning, crop selection, varietal selection, production practices (which include among other irrigation, cultural practices, fertilization and chemical treatments) and -quality attributes to satisfy market and consumer requirements. The overall quality of fresh produce cannot be improved after harvest, there are a few exceptions like controlled ripening to improve color and flavor and refrigerated and controlled cold storage to extend shelf-life. These exceptions are however limited by pre-harvest conditions.

A. **Size:** It is the major quality factor and farmers through varietal selection followed by production practices, should try to control it. The aim is to produce a crop of an average desirable size which is not necessarily the largest possible size.
B. **Mechanical defects:** The need to reduce surface defects is an inevitable consequence of market development. These defects can be physical damages which can be controlled pre-harvest by better plant management to protect the produce from wind, sun, hail, chemical residues, etc.
C. **Pest and diseases:** A better control with appropriate and controlled application of chemicals and field sanitations practices to control fungal, bacteria and insect attacks and avoid the presence of unsightly marks.

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D. **Varietal selection:** The desired market characteristics of a commodity can often only be obtained by changing the variety being grown. This will often mean that growers have the difficult task of adapting to new varieties which may have very different growing characteristics compared to traditional ones. However, a resistance to change may mean a major loss of income as the traditional varieties become less popular on the market.

Horticultural crop production involves a series of intricate processes, among which harvesting and post-harvest handling are pivotal for maintaining product quality and marketability. Harvesting marks, the culmination of months of cultivation efforts and represents a critical juncture where improper techniques can lead to significant losses. Efficient harvesting involves identifying optimal maturity, careful detachment, and collection of fruits to prevent damage. Post-harvest handling encompasses a range of activities aimed at preserving the quality of freshly harvested produce. Key factors influencing post-harvest quality include maturity stage, harvesting techniques, precooling, sorting, grading, packaging, storage, and transportation. Improper handling at any of these stages can accelerate deterioration, leading to losses in yield and quality.

The quality and conditions of produce sent to market and their consequent selling price are directly affected by the care taken during harvesting and field handling. Whatever the scale of operations or the resources of labor and equipment available, the planning and carrying out of harvesting operations must observe basic principles. The objective of the grower should be: - to harvest a good quality crop in good condition; - to keep the harvested produce in good condition (protected from rain, sun or animals) until it is consumed or sold; and - to dispose of the crop to a buyer or through a market as soon as possible after harvest. To meet these objectives, success for the harvesting, field handling and marketing must depend on planning from the earliest stages of production, with particular regard to: - crop selection and timing to meet expected market requirements; - contact with buyers so that the crop can be sold at a good price when ready for harvest; - planning harvest operations in good time, arranging for labor, equipment, material, cover space to protect the inputs and the harvested crop and transport; - it is economically sound in terms of returns on investment to improve grading, packing and handling of the produce before it leaves the farm; - considering the above the grower must ensure that all those working on the farm are properly trained; - provide full supervision at all stages of harvesting and field handling.

Maturity and ripening

Harvest maturity is one of the most important factors that influence postharvest life and the final quality of products, such as appearance, texture, taste, flavor, and nutritional value. The word "maturity" means "fully developed." The word “mature” is derived from the Latin word "maturus" meaning “maturation.” When the maturation is completed, fruit reaches the harvesting stage. In general, maturation can be accomplished when the crop is on the tree or on the plant, and in some cases, maturation can continue even after the fruit has been detached from the tree or plant. In general, as fruit ripens, quality attributes such as color and flavor increase, but storability declines (Figure 1). For this reason, it is crucial to determine the correct harvest maturity stage in such fruits that can be ripened after harvest.

Maturation is a sign of fruit ready to harvest. At this point the edible part of the fruit or vegetable is fully developed in shape and size, even though it may not be ready for immediate consumption. In most cases, ripening follows or overlaps with maturation and the produce becomes edible. The term “maturation” has different stages, including immature, mature, fully mature, and over mature. The quality and nutrient status of horticultural crops vary depending on their maturity levels. For example, anthocyanin concentrations of fully matured red strawberries are usually higher than that of white tip (immature) strawberries, while total flavonoids, phenolics, and total antioxidant activities of fruit harvested at the white tip stage are usually greater than those harvested at the red ripe stage. In addition, the harvest of immature fruits will result in reduced size and yield, poor quality, and uneven ripening. Generally, a compromise between an earlier and a late harvest has to be reached to achieve the premium quality for the consumer and in the same time extend postharvest life for marketing.

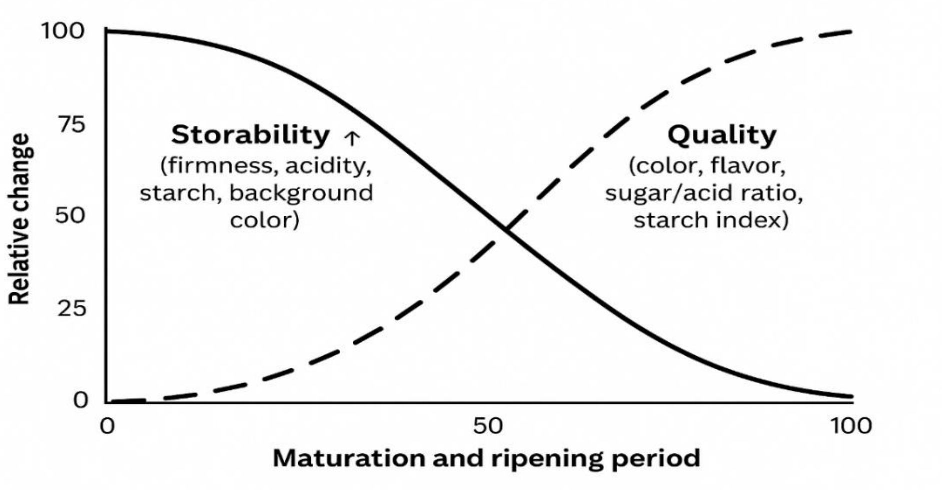

Figure 1: The relationship between the storability and quality in fruit and vegetables

The word "ripe" is derived from Saxon word "ripi", which means "to gather or reap." This is the condition of maximum edible quality attained by the fruit. Ripening is a consequence of biochemical processes, which transform a physiologically mature but inedible fruit into an edible one. Ripening involves a series of changes occurring during the early stages of senescence, in which structure and composition of unripe fruit is so altered that it becomes acceptable to eat. Ripening is a process resulting in softening, coloring, and sweetening as well as an increase in aroma compounds that make fruit ready to be consumed or processed. Some fruits need to ripen for consumer acceptance after harvest. Senescence is the last stage of postharvest phase before death, and it is characterized by natural degradation, loss of texture, flavor, and other quality parameters. The maturity of a fruit/vegetable can be considered in two different parts: physiological maturity and horticultural maturity.

A. **Physiological maturity:** refers to the stage of development where a fruit/vegetable can continue with the developmental processes even after being detached from the parent plant.
B. **Horticultural maturity:** refers to the stage of development in which a fruit/vegetable possesses the prerequisites for use by consumers. It is sometimes referred to as commercial maturity. The maturity of a fruit or vegetable at harvest, whether physiological or horticultural, is a very important factor that determines the storage life as well as the final quality of the product.

Determining maturity: The common methods of determining the maturity of a fruit are: 1. Visual method: this method largely depends on the outward appearance of the fruit, like skin color, size (mass and volume), presence of dry mature leaves, fullness of fruit, drying of plant body. 2. Physical method: firmness, specific gravity, ease of separation of abscission, etc. 3.

Chemical method: total soluble solids (TSS), titratable acidity (TA), TSS:TA ratio, starch content, and so on. 4. Computational method: days from flower bloom or bud initiation stage until optimum fruit development. 5. Physiological method: respiration, aroma development, and so on.

Maturity at harvest: Fruits and vegetables are considered to be commercially mature when at the stage of physiological development that consumers consider to be the most desirable. However, the stage at which a commodity reaches commercial maturity, varies greatly. Climacteric fruits are generally harvested prior to the onset of natural ripening so they can be transported in a hard-green stage and then ripened in the market. Some other fruits and vegetables are harvested slightly immature as they are then less susceptible to attack by microorganisms. For details on maturity indices see the following Tables 1 and 2.

Maturity standards have been determined for many fruits and vegetables. Harvesting crops at the proper maturity allows handlers to begin their work with the best possible quality produce. Produce harvested too early may lack flavor and may not ripen properly, while produce harvested too late may be fibrous or overripe and have a shorter shelf-life. Pickers can be trained in methods of identifying produce that are ready for harvest.

Table 1. Maturity indices for fruits

Maturity index	Examples
Elapsed days from full bloom to harvest	Apples, pears
Mean heat units during development	Peas, apples, sweet corn
Development of abscission layer	Some melons, apples, feijoas
Surface morphology and structure	Cuticle formation on grapes, tomatoes, Netting of some melons, Gloss of some fruits (development of wax)
Size	All fruits and many vegetables
Specific gravity	Cherries, watermelons, potatoes
Shape	Angularity of banana fingers, full cheeks of mangoes, compactness of broccoli and cauliflower
Solidity	Lettuce, cabbage, Brussels sprouts
Firmness	Apples, pears, stone fruits
Tenderness	Peas
Color (external and internal color)	All fruits and most vegetables Flesh color of some fruits, formation of jelly-like material in tomato fruits
Starch content	Apples, pears
Sugar content	Apples, pears, stone fruits, grapes
Acid content, sugar/acid ratio	Pomegranates, citrus, papaya, melons, kiwifruit
Juice content	Citrus fruits
Oil content	Avocados, Persimmons, dates
Astringency (tannin content)	Apples, pears

Harvesting

Harvesting plays a crucial role in horticultural crop production and any deficiencies during this phase can result in the loss of an entire year's effort. Utilizing improper harvesting techniques can lead to crop damage through bruising, which may occur due to compression (resulting from overfilling containers or storage facilities), impact (caused by dropping or external objects striking the crop), or vibration (due to inadequate packing during transportation). It is the removal of fruits from the tree (Prasad et al., 2018). Generally, fruits are harvested when they have attained proper size, attractive color and acceptable quality (i.e proper maturity). Harvesting is one of the important operations, that decide the quality as well as storage life of produce and helps in preventing huge losses of fruits (Rathore et al., 2012). Harvesting of fruits should be done at optimum stage of maturity. During harvesting operation, a high standard of field hygiene should be maintained. It should be done carefully at proper time without damaging the fruits. The harvesting operation includes:

- Identification and judging the maturity of fruits
- Selection of mature fruits
- Detaching or separating of the fruits from the tree and
- Collection of matured fruits.

Table 2. Maturity indices for vegetables and root crops

Crop	Index
Root, bulb and tuber crops	
Radish and carrot	Large enough and crispy (over-mature if pithy)
Potato, onion, and garlic	Tops beginning to dry out and topple down
Yam, bean and ginger	Large enough (over-mature if tough and fibrous)
Green onion	Leaves at their broadest and longest
Fruit vegetables	
Cowpea, yard-long bean, snap bean, sweet pea, and winged bean Lima bean and pigeon pea	Well-filled pods that snap readily Well-filled pods that are beginning to lose their greenness
Okra	Desirable size reached and the tips of which can be snapped readily
Snake gourd, and dishrag gourd	Desirable size reached and thumbnail can still penetrate flesh readily (over-mature if thumbnail cannot penetrate flesh readily)
Eggplant, bitter gourd, chistophine or slicing cucumber	Desirable size reached but still tender, over mature when color changes dull and seeds are tough Seeds slipping when fruit is cut, or green color turning pink
Tomato	Exudes milky sap when thumbnail penetrates kernel Deep green color turning dull or red
Sweet corn	Easily separated from vine with a slight twist leaving clean cavity
Sweet pepper	Change in fruit color from a slight greenish white to cream, aroma noticeable
Musk melon, Honeydew melon, Water melon	Color of lower part turning creamy yellow, dull hollow sound when thumped
Flower vegetables	
Cauliflower	Curd compact (over mature if flower cluster elongates and become loose)
Broccoli	Bud cluster compact (over mature if loose)
Leafy vegetables	
Lettuce	Big enough before flowering
Cabbage	Head compact (over mature if head cracks)
Celery	Big enough before it becomes pithy

i. **Harvesting time:** When the decision to harvest has been taken the preferred time for harvest varies from crop to crop. However, the preferred time is the coolest part of the day, usually in the early morning or late afternoon. This is particularly so for leafy vegetables. Other factors such as the availability of labor and transport and the distance to a packing house or temporary storage area may dictate that some other harvesting time is more suitable or necessary. The selected time should be that which minimize the time between harvest and transport to a packing shade. For example, if night transportation is used it is not advisable to harvest early morning unless produce can be placed under cover and a well-ventilated place during the daytime. Local weather conditions could affect the harvesting time:
   - ❖ It is not desirable to harvest produce when it is wet from dew or rain as this greatly increases the risk of post-harvest spoilage and the tissue is more prone to physical damage.
   - ❖ It is also not advisable to harvest during a hot and sunny day if the produce is left in the field and cannot be protected by the sunrays and the heat.
   - ❖ Protect harvested produce in the field by putting it under open-sided shade when transport is not immediately available. Produce left exposed to tropical sunlight will get very hot. Leafy green vegetables such as spinach and salad lose water quickly because they have a thin waxy skin with many pores.

Fruits and vegetables are living entities and therefore, should be harvested with great care. Harvesting practices should cause as little mechanical damage to produce as possible. The following points should be kept in mind while harvesting the crop: ¾ Gentle picking and harvesting will help reduce crop losses. ¾ Wearing cotton gloves, trimming finger nails, and removing jewelry such as rings and bracelets can help reduce mechanical damage during harvest. ¾ Produce should be harvested during coolest part of the day not wet from dew or rain. ¾ Empty picking containers with care. ¾ Keep produce cool after harvest (provide shade).

ii. **Harvesting factors:** The quality and conditions of produce sent to market and their consequent selling price are directly affected by the care taken during harvesting and field handling. Whatever the scale of operations or the resources of labor and equipment available, the planning and carrying out of harvesting operations must observe basic principles. The objective of the grower should be: - to harvest a good quality crop in good condition; - to keep the harvested produce in good condition (protected from rain, sun or animals) until it is consumed or sold; and - to dispose of the crop to a buyer or through a market as soon as possible after harvest.

To meet these objectives, success for the harvesting, field handling and marketing must depend on planning from the earliest stages of production, with particular regard to:

- ✓ crop selection and timing to meet expected market requirements;
- ✓ contact with buyers so that the crop can be sold at a good price when ready for harvest;
- ✓ planning harvest operations in good time, arranging for labor, equipment, material, cover space to protect the inputs and the harvested crop and transport;
- ✓ it is economically sound in terms of returns on investment to improve grading, packing and handling of the produce before it leaves the farm;
- ✓ considering the above the grower must ensure that all those working on the farm are properly trained;
- ✓ provide full supervision at all stages of harvesting and field handling.

iii. Methods of harvesting

a. **Manual harvesting (Hand harvesting):** Harvesting by one's own hand is called manual harvesting. It is done in soft fruits such as strawberries and raspberries which are borne in low growing points. Fruits, which are borne on trees, such as citrus, mangoes, pears, peaches etc. are difficult to harvest. It is carried out in several ways. Harvesting can take quite a long time.

b. **Mechanical harvesting:** A number of semi/ automatic mechanical devices are used for harvesting the produce on commercial scale. The fruits required for processing may be harvested mechanically for e.g orange for juice making.

c. **Chemical harvesting:** Some chemical sprays are used in certain fruit crops for the formation of natural abscission layer on the fruit stalk and should be applied prior to fruit harvesting ethephon, abscissic acid and cycloheximide proved to be effective.

iv. Modern Harvesting Techniques

- **Picking bag:** Cloth bag with openings on both ends can be easily worn over the shoulders with adjustable harnesses. In case metallic buckets are to be used for harvesting, fitting cloth over the opened bottom can reduce damage to crop. Fitting canvas bags with adjustable harnesses, or by simply adding some carrying straps to baskets also helps to reduce handling losses.
- **Tripod ladders:** A ladder with three legs is very convenient and more stable than a common ladder. A ladder helps harvesting crops such as mango, kinnow, pears, peaches, plums without damaging tree branches.
- **Picking poles and catching sacks:** These tools can be easily made by hand. A long pole attached to a collection bag, allow the harvester to cut catch produce growing on a tree without climbing on tree. The collection bags can be hand woven from strong cord or sewn from canvas. The hoop used as the collection bag rim and sharp cutting edges can be made from sheet metal, steel tubing or recycled scrap metal.
- **Clippers and Knives:** Some fruits such as citrus, grapes and mangoes, need to be clipped or cut from the plant. Clippers or knives should be kept well sharpened and clean. Peduncles, woody stems or spurs should be trimmed as close as possible to prevent fruit from damaging neighboring fruits during transport. Care should be taken to harvest pears so that the spurs are not damaged. Pruning shears can be used for harvesting fruits and some vegetables.
- There are different types of automatic mechanical harvesting methods, such as limb shaking, canopy shaking, trunk shaking, air blasting, and robotic harvesting.
- **Limb shakers:** These are used to harvest the fruits for processing. For example, citrus fruits, apricots, prunes, peaches and sour cherries can be harvested by using limb shakers. Shakers can be controlled remotely from the operator's handle on the shaker. These limb shakers are effective in removing a high percentage of fruit by imparting long strokes to limbs at a low frequency.
- **Canopy shaker:** In this system, citrus fruits are harvested by the vibrating mechanism causing the tines to impact fruit directly or by impacting fruit-bearing branches. The canopy shaker can be used to clamp secondary limbs to shake vertically to pick the fruit.
- **Trunk shakers:** These are used to remove fruit, mainly deciduous fruits, olives, nuts, and citrus fruits. Generally, a tractor-mounted trunk shaker is used on cultivars of different fruits in comparison to a hand-held shaker. Overall the tractor-mounted

shaker is more effective, with about a 72% detachment than the hand-held shaker with a 57% detachment. However, in this system, defoliation risk can be a problem at a high-shaking frequency. Furthermore, trunk shakers cause certain problems by damaging trees resulting in more susceptibility for fungal attacks. The removal rate of trunk shakers varies from 67% on large trees to 98% on small trees. For example, the percentage of fruit detached by the trunk shaker ranges between 70% and 85% on mandarins and oranges in Spain.

- **Air blast:** A force-generated air blast may be used to remove the fruit from the tree. In this system, oscillating air blast machines are used, and the fruit detachment rate is maximized by the oscillation rate. The performance of an air blast harvester is dependent on various factors, such as the tree structure, size, weight of the fruit, and fruit load of trees.
- **Robotic harvester:** Despite the high degree of mechanization and automation in agriculture, very few robotic harvesters have evolved beyond the research stage, and none are widely used in open fields. The limited success of robotic harvesters is mainly due to the complexity of the terrain, environment, and mission, which results in low fruit-picking success ratios or operations that are too slow to be economically relevant. Research on robotic harvesters is ongoing.

Post-harvest handling/packhouse operations

Fruits and vegetables, being living organs, continue to respire even after harvesting when their food reserves are limited. Alongside the degradation of respiratory substrates, various changes occur in taste, color, flavor, texture, and appearance in harvested commodities, rendering them unsuitable for consumption if not handled properly. Improper temperatures can disrupt the normal metabolism of harvested organs. Higher temperatures can significantly accelerate metabolic activities, reducing shelf life, while much lower temperatures may cause freezing or chilling injuries in the harvested produce. Fruits and vegetables that are mechanically damaged during harvesting are highly susceptible to fungal decay during storage. Infected produce may also spread diseases to adjacent stored commodities if not sorted out before storage. Postharvest treatments play a crucial role in maintaining the external quality of fruits, ensuring better prices and returns for farmers. It is widely acknowledged that while the quality of harvested commodities cannot be further improved, it can be preserved until consumption by reducing the rate of metabolic activities through appropriate postharvest treatments and handling operations (Prasad et al., 2016; Rathore et al., 2012; Thompson, 1996).

The actual operations to be carried out will be determined by the market system and volume of produce passing through the packing place. A packinghouse can fulfill a variety of functions. It may, for example, accept produce from the field and prepare it for the market or act as a preliminary quality control operation to ensure that only quality produce is placed into storage. A packing house may also be used to repack produce after storage for retail marketing. Packing houses for small-scale operations can be utilized for:

- Receiving produce where, the quantity and quality are checked on arrival before being transferred to a temporary storage area;
- Treatment, where excess part plants are trimmed off, surface contaminants removed by washing or brushing, and chemicals applied to extend storage or market life;
- Grading, where substandard items are removed, produce is graded for factors such as maturity, color and the separated grades are sized and packed; and
- Temporary shelter for produce waiting for being loaded for dispatch.

Selection of Fruits and Vegetables

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Dumping

(Unloading onto Conveyor)

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Washing

(Removal of Dirt & Contaminants)

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Drying & Brushing

(Water Removal & Surface Cleaning)

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Wax Application & Shining

(Enhancing Appearance & Shelf Life)

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Grading

(Sorting by Size, Colour, Ripeness, and Quality)

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Sizing

(Classifying into Standard Categories)

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Boxing

(Packing into Crates, Cartons, or Boxes)

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Sealing

(Labeling & Quality Assurance)

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Shipment

(Local Market (Fresh Consumption) and Processing Facilities (Juicing, Canning, Freezing)

Figure 3: Post harvest handling operations in a packhouse

All packing house operations can be executed manually, mostly small scale, or with a range of mechanical devices of varying sophistication, mostly for larger operations. For packinghouses with small throughput, economics will favor manual operations with workers performing more or all the tasks. The advantage of human labor is that it provides flexibility of operation since it can perform a variety of tasks with equal efficiency.

Dumping: The first step of handling is known as dumping. It should be done gently either using water or dry dumping. Wet dumping can be done by immersing the produce in water. It reduces mechanical injury, bruising, abrasions on the fruits, since water is gentle on produce. The dry dumping is done by soft brushes fitted on the sloped ramp or moving conveyor belts. It will help in removing dust and dirt on the fruits.

Pre-sorting: It is done to remove injured, decayed, mis-shapen fruits. It will save energy and money because culls will not be handled, cooled, packed or transported. Removing decaying fruits are especially important, because these will limit the spread of infection to other healthy fruits during handling.

Pre-cooling: The practice described above is considered obligatory in developed countries for nearly all perishable commodities. This procedure, known scientifically as precooling, involves rapidly lowering the temperature of freshly harvested agricultural products from their initial field temperature (referred to as pulp temperature) to the optimal storage temperature (Brosnan and Sun 2001). Precooling is of paramount importance as it greatly extends the shelf life of agricultural produce. The removal of field heat has been found to profoundly affect the rate of respiration and all biochemical processes occurring within newly harvested crops. Fruits, vegetables, and flowers continue to respire even after harvest, leading to degradation such as loss of nutrients, changes in texture and flavor, and mass reduction (Sargent et al., 1988). While these processes are irreversible, their progression can be significantly slowed down through precooling techniques before storage or distribution. Four primary precooling methods-forced air cooling, hydro cooling, vacuum cooling, and icing-are applied based on the inherent texture and commercial value of the produce (Ahmad and Siddiqui 2015). Each method is tailored to specific crops. The rate of precooling is contingent upon several factors:

- The temperature differential between the crop and the cooling medium.
- The accessibility of the cooling medium to the crop.
- The characteristics of the cooling medium itself.
- The velocity or speed of the cooling medium.
- The efficiency of heat transfers from the crop to the cooling medium.

a. **Forced air cooling:** Forced air cooling stands out as the premier precooling method for fruits, employed worldwide to combat field heat, disease progression, softening, and weight loss. This technique swiftly extracts field heat by circulating cold air over the produce. Fans located within a refrigerated storage space pull air from the crates holding the produce into the cooling unit. Forced air cooling proves highly effective across a range of commodities, particularly excelling in the cooling of berries and stone fruits, which are especially receptive to this specific cooling approach (Thompson et al., 2008).

b. **Vacuum cooling:** Produce is placed within a tightly sealed chamber where atmospheric pressure is substantially lowered. In this reduced pressure environment, a phenomenon similar to boiling occurs within the produce, facilitated by its internal thermal energy. This process causes some of the water within the produce to vaporize, thereby reducing the overall temperature of the produce. Heat and moisture are removed from the vacuum chamber using mechanical refrigeration (McDonald and Sun 2000).

c. **Hydro cooling:** Hydro cooling employs chilled water to reduce the temperature of perishable agricultural goods. Cooling packed fruits with this method poses certain challenges. Typically, mechanical refrigeration is used to cool the water, although alternatives such as cold well water and ice are sometimes utilized. The size of hydro cooling units varies based on the scale of operation, but significant refrigeration or ample ice is essential to maintain the water within the target temperature range of 33–36 °F. The produce is cooled through immersion in a water bath or via a sprinkler system. A wide range of fruits that can tolerate moisture exposure can undergo hydro cooling (Teruel et al., 2004).

d. **Icing:** Crushed or slurry ice is directly added to the containers holding perishable agricultural goods. This method has proven effective in the precooling of certain vegetable containers. The produce can rapidly cool within a short timeframe, thereby helping to maintain its temperature during transportation (El-Ramady et al., 2015).

Sorting and grading: This process is primarily carried out to ensure the high standard of packaging and to remove produce afflicted with diseases or defects from the batch. Employing suitable sorting and grading methodologies instills confidence in the quality of agricultural products. Typically, this activity takes place either in the agricultural field or within designated storage facilities (Atkari et al., 2013). Both manual and mechanical graders are utilized for grading purposes, with mechanical graders efficiently grading fruits and vegetables with a spherical shape. Grading involves assessing attributes such as color, size, and the extent of imperfections. Sorting, however, relies entirely on human labor to eliminate fruits or vegetables affected by diseases, defects, or damage.

Washing and cleaning: In India, fruit washing is not commonly practiced among farmers, but fruit business enterprises often establish washing facilities within their pack houses or cold storage facilities. Washing may not be considered necessary for certain fruits like grapes and litchi. However, for fruits such as apple, plum, and guava, thorough washing before storage is highly recommended. Washing grapes removes natural wax, while introducing browning in litchi fruit enhances their shelf life and visual appeal. Washing pome fruits before storage, particularly in Controlled Atmosphere Storage Chambers (CASC), has been found beneficial. This practice increases humidity levels within the storage chamber, thereby reducing spoilage occurrences (Ozden and Bayindirli 2002).

Postharvest treatments: It was observed that the presence of 1-methyl cyclopropene had an inhibitory effect on the production and/or action of ethylene in fruits during the processes of ripening and storage (Yuan et al., 2010). It has been reported that gibberellic acid (GA), kinetin, and silver nitrate can slow down respiration rates of fruits.

a. **Thermal treatment**: Thermal treatments, which may include hot water, vapor heat, or hot water rinse brushing, are widely used non-chemical methods for reducing postharvest deterioration and insect infestation in many fruits.
b. **Hot water treatment:** Fruits can undergo immersion in heated water to manage various postharvest pathogens (such as larvae and inoculums) and enhance pigmentation of the fruit peel. For mangoes, it's recommended to subject them to a high-water temperature (HWT) of 50-52 °C for 5 minutes. This thermal treatment effectively eliminates fruit fly larvae and helps control microbial infections that may occur during marketing. Additionally, this treatment promotes uniform ripening within 5-7 days (Anwar and Malik 2007).
c. **Vapor heat treatment:** This treatment has shown a high level of effectiveness in controlling fruit fly infections within confined containers (Schirra et al., 2000; Armstrong and Mangan 2007). The boxes are arranged vertically in a room and subjected to an increase in thermal energy and moisture content by injecting steam. Carefully controlled temperature and exposure durations ensure elimination of all developmental phases of insects, including eggs, larvae, pupae, and adults, while preserving the fruit's integrity (Singh and Saini 2014). For citrus, mangoes, papaya, and pineapple, a scientifically recommended treatment involves subjecting them to a temperature of 43°C in an air-saturated environment for 8 hours, followed by maintaining this temperature for an additional 6 hours.
d. **Fumigation:** The use of sulfur dioxide gas (SO_2) in fumigation has been demonstrated as an effective method for controlling postharvest diseases in grapes, especially powdery mildew caused by the pathogen *Botrytis cinerea*.
e. **Waxing:** Waxing fruits is a common practice in the food industry to improve appearance, prolong shelf life, and protect them during transportation. When conducted properly with food-grade waxes like carnauba wax or synthetic waxes, which are edible and safe, it is generally feasible. The edibility of these waxes has been confirmed by experts, suggesting their potential as a viable postharvest

intervention for extending storage duration of various fruits like mango, kinnow, and sweet orange (Abbasi et al., 2011). Waxing offers benefits such as enhanced visual appearance, reduced moisture loss and shrinkage, mitigated postharvest deterioration, and extended storage duration. Paraffin, carnauba, and shellac are commonly used materials, each possessing unique properties that affect lustre, gas permeability, and other physical attributes.

Packaging: Packaging is a blend of artistic, scientific, and technological principles aimed at ensuring safe product transportation to consumers while maintaining optimal condition and minimizing costs. When selecting packaging for fresh produce, preventing physical injury and pressure damage during handling is crucial for achieving optimal shelf-life (Selin 1977). Common packaging materials include wooden crates, jute sacks, polyethylene, High Density Polyethylene, and Cardboard Boxes/CFB (corrugated fiber board) boxes, with various techniques tailored to the nature of the produce (Thompson 1996).

Storage: Fruits are subjected to seasonal variations in availability, with harvesting occurring annually within specific timeframes. Despite this, there is a year-round demand for fruits like guava, apple, mango, grapes, and others. Meeting this demand requires appropriate storage of fruits during the harvesting season, followed by sales during the off-season. While storage cannot enhance fruit quality, it can maintain or slow down the rate of quality deterioration within a set timeframe (Siddiqui et al., 2015). The monetary value of fruits tends to increase after storage, emphasizing the importance of storing only high-quality fruits capable of extended storage. Meticulous regulation of temperature and relative humidity within the storage chamber is crucial for successful storage (Siddiqui and Dhua 2010).

Harvesting → Precooling → Sorting → Washing → Waxing/Chemical Treatments → Sizing → Packaging → Storage → Transportation → Wholesaler → Restoring, Resizing, and Repackaging → Transportation → Retailer → Consumer

Figure 4: Post harvest handling operations of fruits in general

Harvesting → Precooling → Cleaning → Trimming → Grading → Sorting → Curing → Sizing → Waxing → Packaging → Storage → Transportation → Wholesaler → Transportation → Retailer → Consumer

Figure 5: Post harvest handling operations of vegetables (B) in general

Transportation: Fresh horticultural produce undergoes transportation through two distinct methods within a given state, while between states, it is subjected to transportation through three distinct methods. Within states, transportation primarily relies on utility vehicles with a load capacity ranging from 1 to 6 metric tonnes, as well as trucks with a load capacity ranging from 8 to 16 metric tonnes. For inter-state travel, trucks, trains, and airplanes are commonly employed modes of transportation. Among these, trucks with a weight capacity of 8 to 16 metric tonnes are the predominant mode, followed by trains, with air transport being the least utilized (Ahmad and Siddiqui 2015). Fresh produce is mainly transported between states using non-refrigerated trucks, as shipping in India holds minimal significance for goods transportation. The Reefer van's operational capacity is limited to the summer months when it is utilized for transporting high-value domestic produce and imported fruits. Both the operation of the Reefer van and the demand for fresh produce are currently experiencing an upward trend.

Ten Important Guidelines for Postharvest Handling in General

1. **Maturity.** Harvest the product at the correct stage of maturity.

2. **Reduce Injuries.** Reduce the physical handling to a minimum; every time the product is handled, it is damaged.
3. **Protect Product**. Protect the harvested product from the sun; bring it rapidly from the field/exposed area to the packing station and keep out of the direct sun. Transport carefully.
4. **Cleanliness & Sanitation**. Keep the packing line as simple as possible and keep it clean. If water is used, use clean water or a sanitizer if the water is reused. Maintain strict worker hygiene.
5. **Pack Carefully.** Sort, classify and pack the product carefully to achieve uniformity and to prevent damage (compression, scrapes, etc.) which causes decay and inferior quality; use an adequate box or container. Packaging can also be informative.
6. **Palletize.** Ensure that the boxes are well placed on the pallet and that the pallet is strapped.
7. **Cool.** Cool the product as soon as possible after harvest; generally, for every hour of delay from harvest to initiate cooling, one day of shelf-life is lost. Lowering product temperature is the most important way to reduce deterioration.
8. **Know Product**. Know the requirements of the market (size, ripeness, etc) and the product handling requirements (temp., RH, shelf-life, etc.) of the product.
9. **Coordination**. Always try to coordinate the postharvest handling so that it is efficient and rapid. Postharvest handling maintains the quality of a product, it cannot improve it.
10. **Training.** Train and compensate well the workers involved in critical postharvest handling steps; make sure that workers have the necessary tools to facilitate their work.

Conclusion

Pre-harvest factors have profound influence on post-harvest quality and nutritional composition of horticultural crops. Quality deterioration of produce starts right after harvest and continues till consumed. The success or failure of any business plan associated to horticultural commodity solely depends on management practices of factors affecting its quality. The quality of any produce cannot be improved by the use of any post-harvest treatment methods or handling practices but can be maintained. These pre-harvest factors during production should be managed properly for quality production. In order to control the problems of factors affecting post-harvest quality of horticultural produces, education should be provided to all the farmers, laborers, and merchants about basic science and appropriate method of handling at all post-harvest stages could significantly reduce losses in post-harvest chain.

In developing country like India minimization of post-harvest losses is beneficial to combat the growing hungry population in coming days. Proper storage infrastructure, transport facility, involvement of educated and skilled labor is mandatory in loss reduction. Market strategy and consumer awareness regarding the losses should be acknowledged. Most importantly, appropriate government policies and regulations should be implemented for the further development. Future research in these factors which are under human control including field management practices, cultivar selection and improvement offers great potential for enhancing post-harvest quality of horticultural crops. All together reduce the losses in perishable commodities like fruits and vegetables will boost the economy in sustainable way.

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